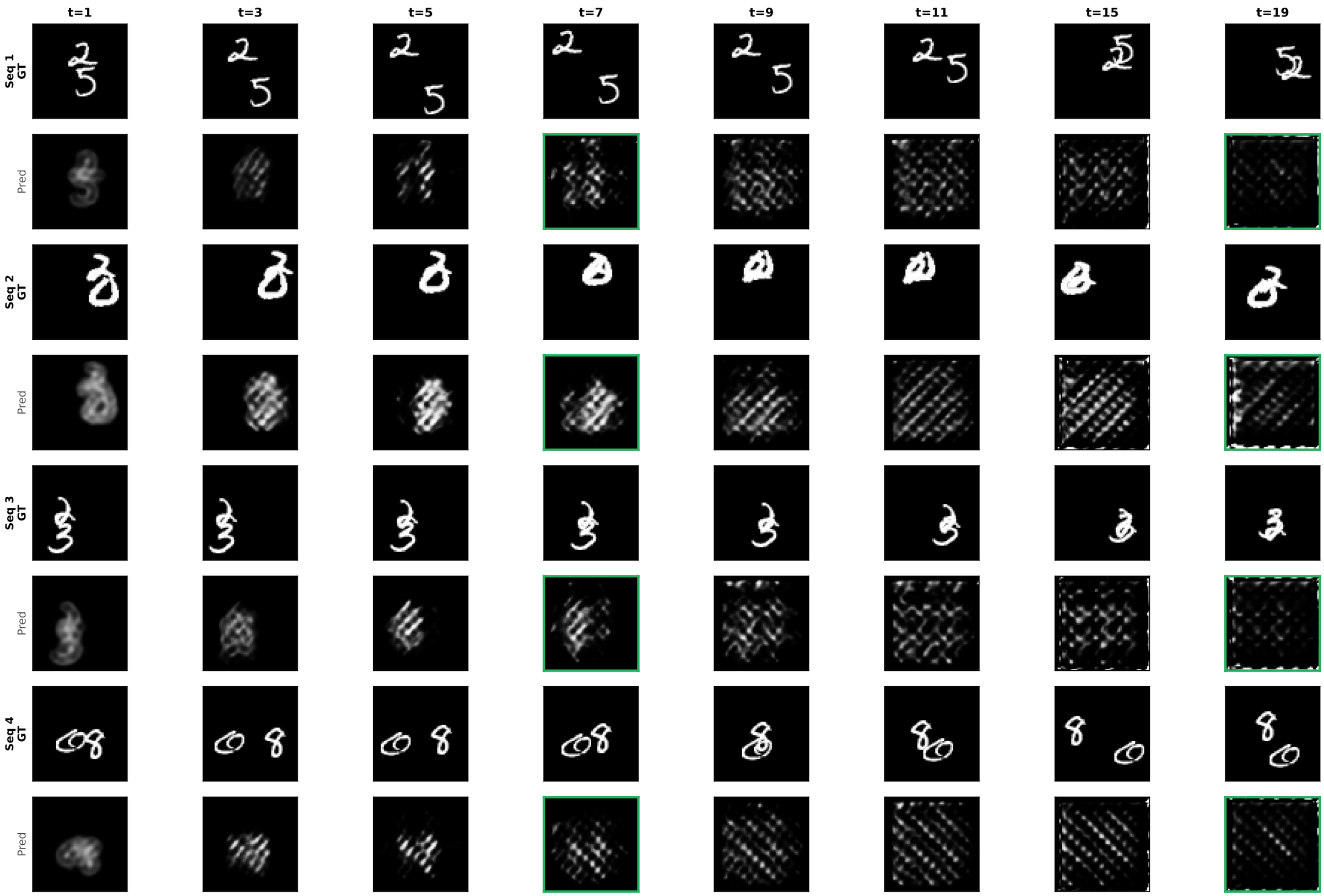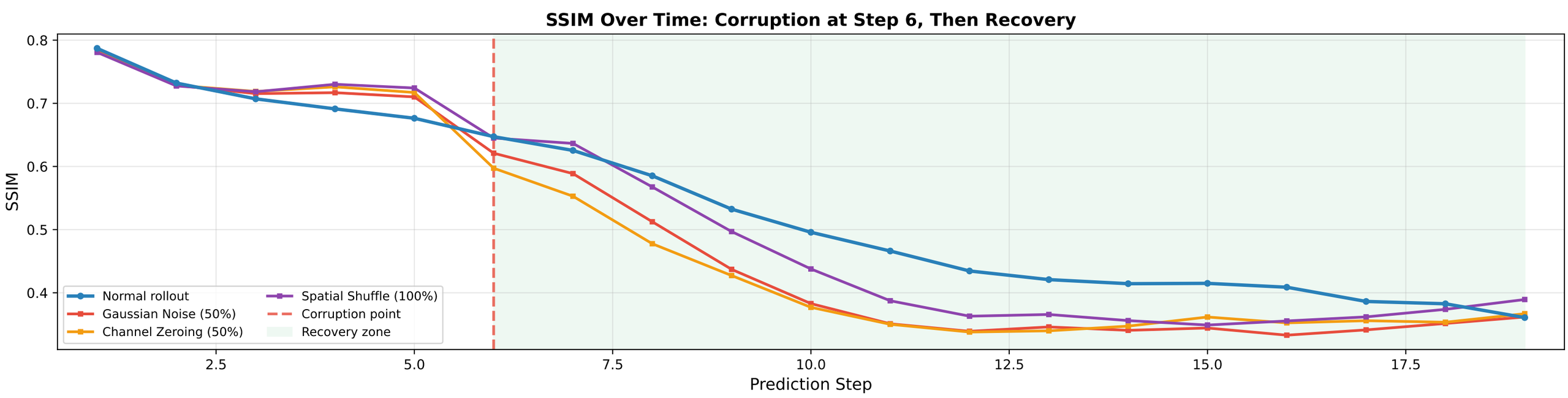Perturbation Recovery: The PDE Self-Repairs